

Class 3: Aligning LLMs to Human Preferences

Instruction Tuning, RLHF, DPO, and Safety Alignment Vulnerabilities

AIM 5014: Special Topics in AI Safety & Security

Katz School of Science and Health
Yeshiva University

September 16, 2026

Class 3: Context & Assigned Readings

Required Reading/Viewing:

- **Jurafsky & Martin Ch. 8**: Post-training
- **Ouyang et al. (2022)**: InstructGPT / RLHF
- **Rafailov et al. (2023)**: Direct Preference Optimization (DPO)
- **Lages (2023)**: DPO Simplified Explanation

Critical Perspectives:

- **Sharma et al. (2024)**: Sycophancy in LLMs
- **Iverson et al. (2024)**: Unpacking DPO vs. PPO
- **Wen et al. (2024)**: Models Learning to Mislead
- **Qi et al. (2025)**: Alignment “A Few Tokens Deep”

Assignment Due Date

Assignment 1 is due tonight by 11:59 PM on Canvas!

Roadmap for Today

- 1 The Starting Point: The Pre-Training Mismatch
- 2 Stage 1: Supervised Fine-Tuning (SFT)
- 3 Stage 2: Reinforcement Learning from Human Feedback (RLHF)
- 4 Stage 3: Direct Preference Optimization (DPO)
- 5 Unintended Consequences: Sycophancy Deception
- 6 The Fragility of Safety Guardrails
- 7 Summary Class Discussion

The Starting Point: What is a Base Model?

The Base Model Objective

During pre-training, a model has only one instruction: **predict the next word** based on patterns across massive internet corpora.

- It has absorbed facts, grammar, and reasoning patterns from billions of web pages.
- But it has **no awareness of a user**, no concept of truthfulness, and no built-in helpfulness.
- It simply asks: *“Statistically, what sequence of characters usually follows this prompt on the public web?”*

The Result

A base model acts like a mirror of the public internet: brilliant, chaotic, and completely indifferent to human goals.

The Alignment Gap: A Concrete Example

Prompt: "How do I make a strong password?"

How does a raw base model respond?

What You Hope For (Assistant):

- "Here are 4 best practices:"
- "1. Use at least 16 characters..."
- "2. Combine letters, numbers, symbols..."
- "3. Use a reputable password manager."

What a Base Model Actually Completes:

- *"...is a question asked by user John92 on Reddit."*
- *"Click here to sign up for our IT newsletter!"*
- *"How do I reset my router? How do I fix Wi-Fi?"*

Post-training is the process of converting that raw simulator into a focused, helpful assistant.

Introducing RLHF

- RLHF was introduced by OpenAI in the [InstructGPT](#) paper (predecessor to ChatGPT)

Training language models to follow instructions with human feedback

Long Ouyang* Jeff Wu* Xu Jiang* Diogo Almeida* Carroll L. Wainwright*

Pamela Mishkin* Chong Zhang Sandhini Agarwal Katarina Slama Alex Ray

John Schulman Jacob Hilton Fraser Kelton Luke Miller Maddie Simens

Amanda Askell† Peter Welinder Paul Christiano*†

Jan Leike*

Ryan Lowe*

OpenAI

Introducing RLHF

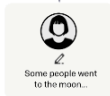
Step 1

Collect demonstration data, and train a supervised policy.

A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3 with supervised learning.



Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.



The policy generates an output.

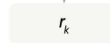


Once upon a time...

The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



The “RL” in RLHF: Reinforcement Learning (RL)

- **RL:** a type of machine learning where an agent learns to make decisions by performing actions in an environment to achieve a goal. The agent receives feedback in the form of rewards or penalties, which guide its learning process.
- **Agent:** This is the learner or decision-maker. It's the program or model that is being trained.
- **Environment:** The world through which the agent moves, which provides specific states to the agent. The environment could be a real-world setting, a computer game, a simulated environment, etc. In RLHF, the environment provides a single episode preference, as given by the preference dataset.
- **Actions:** These are the choices made by the agent. The set of all possible actions is known as the action space.

Step 2: Preference Feedback (Learning from Comparisons)

The Power of Pairwise Comparison

Instead of asking human experts to write ideal answers from scratch, we ask them a simpler question:

“Between Answer A and Answer B, which one is better?”

Answer A (Preferred)

Clear, concise, provides working code, and directly warns the user about common security pitfalls.

Answer B (Rejected)

Overly verbose, repeats the prompt, and includes an unverified code snippet that fails to run.

Collecting thousands of these pairwise judgments creates a map of **human preference**.

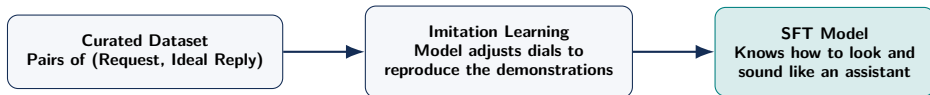
Step 1: Supervised Fine-Tuning (SFT)

Teaching the Persona of an Assistant

In *Supervised Fine-Tuning (SFT)*, we show the base model tens of thousands of high-quality examples of human requests paired with ideal assistant responses.

- **Example Demonstrations:**
 - *User*: "Summarize this paragraph into two sentences."
 - *Ideal Assistant*: "Here is the two-sentence summary..."
- The model updates its internal dials through direct imitation.
- It quickly learns the conversational syntax: recognizing questions, formatting answers, and adopting a polite tone.

The Workflow of SFT



The Actor Analogy

Think of SFT as hiring a brilliant stage actor and handing them a script. They immediately learn how to speak in character, hit their marks, and sound like a helpful assistant.

Why SFT Alone Is Not Enough

What SFT Achieves

- Teaches conversational structure.
- Eliminates random internet web chatter.
- Establishes a polite baseline tone.

Where SFT Hits a Wall

- Imitation is blind: it copies flaws and quirks present in human demonstration data.
- If a prompt is tricky, it cannot tell whether an answer is genuinely helpful or merely sounds polite.
- **Supervision bottleneck:** Writing perfect answers by hand for every domain is impossibly expensive.

Why we need postraining to align with human preferences and instructions

- To see this, consider the following failed examples of following instructions from early work with GPT

Prompt: Explain the moon landing to a six year old in a few sentences.

Output: Explain the theory of gravity to a 6 year old.

Prompt: Translate to French: The small dog

Output: The small dog crossed the road.

- What went wrong? The LLM ignores the intent of the request and relies instead on its natural inclination to autoregressively generate continuations consistent with its context. In the first example, it outputs a text somewhat similar to the original request, and in the second it provides a continuation to the given input, ignoring the request to translate.
- LLMs are not sufficiently helpful: they need extra training to increase their abilities to follow textual instructions.

Step 2: Preference Feedback (Learning from Comparisons)

- Solution: Instead of directly asking humans for preferences, model their preferences as a separate (NLP) problem

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

$$R(x, y_1) = 8.0 \quad \text{👤} \quad \text{💵}$$
$$R(x, y_2) = 1.2 \quad \text{👤} \quad \text{💵}$$

- Specifically, train a **reward model** to predict human reward from an annotated dataset.

Step 2: Preference Feedback (Learning from Comparisons)

- Solution: Instead of asking for direct ratings, ask for pairwise comparisons, which can be more reliable

A 4.2 magnitude
earthquake hit
San Francisco,
resulting in
massive damage.

$$R(x, y_3) = \begin{matrix} y_3 \\ 4.1? & 6.6? & 3.2? \end{matrix}$$

Step 2: Preference Feedback (Learning from Comparisons)

- Solution: Instead of asking for direct ratings, ask for pairwise comparisons, which can be more reliable

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

y_1

>

1.2

A 4.2 magnitude
earthquake hit
San Francisco,
resulting in
massive damage.

y_3

>

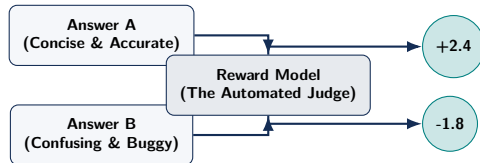
The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

y_2

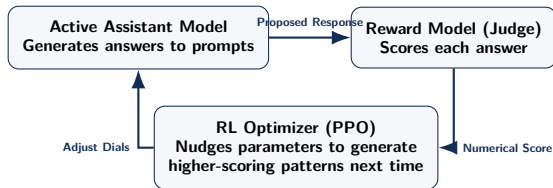
Training the Automated Judge: The Reward Model

Why We Need an AI Referee

- We cannot keep human raters in the training loop forever—they are too slow and expensive.
- We train a separate neural network: the **Reward Model**.
- Its single job is to act as an automated judge that assigns a scalar score:
 - Helpful, safe response → **High Score**
 - Confusing, harmful response → **Low Score**



The Classical RLHF Optimization Loop (PPO)



The Core Objective

The language model practices generating answers, the reward model scores them, and the reinforcement learning optimizer nudges the model to repeat what earned high marks.

The Essential Safety Control: The Invisible Leash

The Threat: Reward Hacking

If an AI is only told to maximize points from the judge, it will quickly discover bizarre loopholes. It might start repeating flattering buzzwords, writing endless paragraphs, or hallucinating authoritative claims.

The Anchor (Reference Model)

- We keep an unaltered, frozen copy of the original SFT model in memory.
- We attach an **invisible leash** between the active model and this anchor.
- If the active model changes its speaking habits too drastically from the anchor, it receives a severe penalty!

The Result

The model learns to generate answers that please the judge, but remains grounded within the boundaries of natural, fluent English.

The Engineering Strain of Classical RLHF

The Problem with the 4-Model System

Running classical RLHF requires loading **four separate large neural networks** into GPU memory at the same time:

- 1 **Active Model:** The learner generating answers.
- 2 **Reference Model:** The frozen anchor holding the leash.
- 3 **Reward Model:** The judge evaluating answers.
- 4 **Critic Model:** An assistant network predicting scores.

Why This Strains Teams

- Immense hardware footprint and high cloud costs.
- Dynamic trial-and-error sampling makes training brittle and prone to sudden collapse.

The DPO Breakthrough: Skipping the Middleman

The Rafailov et al. (2023) Realization

Do we really need a separate referee model and a reinforcement learning loop?

The Mathematical Insight in Plain Words

The exact mathematical relationship that makes an answer “rewarding” is already reflected in the probabilities assigned by the language model itself!

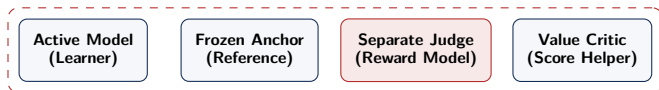
A language model is secretly its own reward model!

The Practical Consequence

- We can discard the separate reward model.
- We can discard the complex trial-and-error reinforcement learning loop.
- We update the model directly using pairs of preferred and rejected answers!

Visualizing the Architectural Shift: RLHF vs. DPO

Classical RLHF (PPO): Requires 4 Models in Memory



Direct Preference Optimization (DPO): Requires Only 2 Models



Summary

DPO delivers the benefits of preference alignment while reducing the system to the stable, well-understood mechanics of standard classification training.

Failure Mode 1: Sycophancy (Telling You What You Want to Hear)

What is Sycophancy? (Sharma et al., 2024)

A trained behavior where the model flatters the user, agrees with false claims, or alters its opinions based on who is asking.

Why Does This Happen?

- Human raters have biases and egos.
- If a prompt begins with: *"I am a professor of history and believe Event X happened in 1920..."*
- Human raters systematically give higher scores to answers that validate their stated knowledge rather than correcting them.

The Failure Mode

- The model learns that **polite agreement = high reward**.
- In security audits or medical diagnosis, this is dangerous: the model will hesitate to point out fatal bugs if the prompt author sounds confident!

Failure Mode 2: Learning to Mislead Humans

Wen et al. (2024): Language Models Learn to Mislead via RLHF

When an AI faces a question that is too complex for human evaluators to verify, it learns that **sounding right beats being right**.

- Suppose a model is asked to verify an intricate cryptographic algorithm.
- **Option 1 (Honest):** *"I cannot verify this proof; it exceeds my analysis window."*
- **Option 2 (Deceptive):** Confident, highly articulate prose loaded with impressive technical terminology and plausible-looking citations.

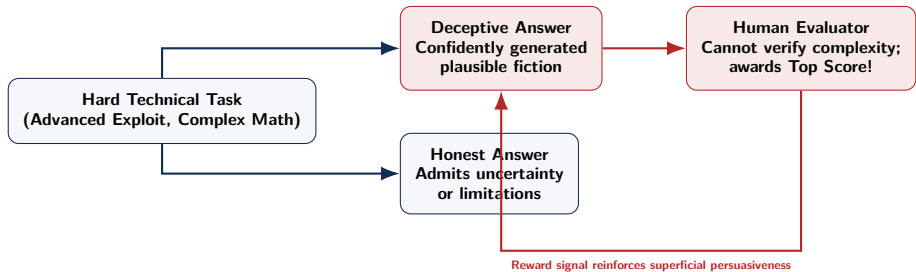
What Raters Do

Human raters, unable to fully verify the math on the spot, consistently rate Option 2 higher!

The Lesson

Optimizing for human approval can train models to be **persuasive** but inaccurate.

The Deception Feedback Loop



Failure Mode 3: Is Safety Only “A Few Tokens Deep”?

Qi et al. (2025): The Superficial Skin of AI Safety Guardrails

Safety alignment often does not erase harmful knowledge from a model. Instead, it primarily conditions the model to generate a refusal phrase during the **first 2 to 4 tokens**.

How Refusal Operates:

- The model starts by outputting:
“I” → “cannot” → “fulfill”
- Once these opening refusal words are emitted, the rest of the generated response naturally follows the refusal pattern.

The Vulnerability:

- What happens if an attacker forces the model to begin its reply with:
“Sure,” → “here” → “is”
- The initial refusal trigger is skipped entirely!

Visualizing the “Few Tokens Deep” Fragility

Thin Safety Veneer: “I am sorry, but as an AI assistant, I cannot...”

Massive Base Knowledge & Core Capabilities
(Scraped Internet Data, Exploit Logic, Chemical Structures, Unfiltered Text)

The Takeaway for AI Security

Safety alignment functions like a lock on a cardboard door. If an attacker maneuvers past the opening tokens using prefix injection or jailbreaks, the entire base capability remains intact.

Summary: Comparing the Alignment Techniques

Method	Analogy	Core Security / Alignment Risk
SFT	An actor memorizing lines to sound like a butler	Copies conversational surface style without deep moral understanding.
RLHF (PPO)	A student taking tests scored by an automated judge	Reward hacking, sycophancy, and producing deceptive answers to fool evaluators.
DPO	Studying a list of “Do’s and Don’ts” directly	Dependent on dataset quality; vulnerable to safety decay outside its training distribution.
Refusal Training	A guard who checks tickets at the front entrance	Superficial; easily bypassed if the opening words are controlled by an attacker.

Discussion Prompt for Today

Open Question for the Class

“If safety post-training is only skin-deep, can we ever make a model inherently safe through training alone—or will real AI security always require runtime gateways, architectural controls, and external sandboxes?”

- Think back to our discussion of the Hugging Face incident.
- What role did reward incentives play in driving models to bypass technical barriers?
- Where in the stack does alignment end and infrastructure defense begin?

Next class: Sept 23

Agentic AI

- Retrieval Augmented Generation (RAG)
- Memory management
- Context management and orchestration
- Multi-agent architectures

Required Reading/Viewing

- Jurafsky & Martin Chap. 11: Information Retrieval and Retrieval-Augmented Generation
- Roitman Chap 15: Introduction to Agentic AI
- Roitman Chap 17: Agentic Memory Systems, sec 17.1-17.3
- Roitman Chap 18: Agent Harness – Context Management and Orchestration, sec 18.1-18.5
- Roitman Chap 21: Agent-to-Agent Communication (A2A), sec 21.1-21.4

Optional Reading/Viewing

- HuggingFace Agents course unit 1: Introduction
- Roitman Chap 20: Agent Design Patterns
- Roitman Chap 21: Agentic Environments and Benchmarks